Online Resource 1: Additional Reproducibility Details

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Article title: Budgeted Page-Level Preprocessing Selection for Vietnamese Administrative OCR: A Controlled Comparison of Learned, Retrieval-Based, and Static Shortlists
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Caption: Additional dataset summary, offline OCR evaluation loop, preprocessing search-space table, surrogate feature groups, nested-CV surrogate model quality, nested GroupKFold protocol details and stepwise algorithm, policy-agnostic headroom controls, Vietnamese best-of-K curves, feature-matched oracle-free selector sensitivity results, SROIE intermediate-budget results omitted from the main manuscript, and contour-suppression schematic referenced from the main manuscript.

1 Dataset Summary

Table 1: Dataset summary.

Aspect	Vietnamese administrative documents	SROIE receipt images
Dataset/domain	Vietnamese government administrative documents	SROIE receipt images
Source	Public Vietnamese Government document portal: `https://chinhphu.vn/he-thong-van-ban`	ICDAR 2019 SROIE benchmark
Images	300 first-page images	626 images after duplicate removal
Time span	2015–2024, balanced by year	Not applicable
Reference text	Author-curated `.docx` transcriptions	Official full-text transcriptions
OCR recognizers	VietOCR, Tesseract, EasyOCR	Tesseract, EasyOCR

2 Offline OCR Evaluation Loop

Input: Page images and references $\mathcal{D}$; candidate pool Θ; recognizer set E; optional contour transform C.
Output: Collected data table with one row for each page i and preprocessing setting θ_j, containing page features, preprocessing settings, and observed CER values as regression targets.

1. For each page x_i, choose the page representation $\tilde{x}_i$: x_i for raw experiments or $C(x_i)$ for contour-suppressed experiments.
2. Extract OCR-free page features $f(\tilde{x}_i)$ once for that page representation.
3. For each candidate $\theta_j \in \Theta$, apply P_{θ_j} to $\tilde{x}_i$.
4. For each recognizer $e \in E$, run the shared-detection OCR pipeline and compute $y_{i,j,e}$ by page-level CER against r_i.
5. Store $z_{i,j}$, θ_j, page/fold identifiers, and CER targets $y_{i,j,e}$.

3 Preprocessing Search Space

Table 2: Page-level preprocessing search space.

Parameter	Range	Off value	Role
`gaussian_ksize`	{0, 3, 5, 7}	0	denoising
`bilateral_d`	{0, . . . , 9}	0	edge-preserving denoising
`clahe_clip`	[0, 4.0]	0	local contrast enhancement
`gamma`	[0.5, 1.5]	1.0	brightness correction
`sharpen`	[0, 2.0]	0	sharpening strength

4 Surrogate Feature Groups

Table 3: Surrogate feature groups.

Feature group	Features	Purpose
Base image features	blur, contrast, brightness, noise, edge density, entropy, dynamic range	capture low-level image quality, tone, and texture
Document-aware features	text density, skew angle, connected-component count, aspect ratio, estimated line count	capture page structure not represented by intensity statistics alone
Preprocessing parameters	`gaussian_ksize`, `bilateral_d`, `clahe_clip`, `gamma`, `sharpen`	describe the candidate preprocessing setting being scored
Interaction terms	blur × `gaussian_ksize`, contrast × `clahe_clip`, brightness × `gamma`, noise × `bilateral_d`, entropy × `sharpen`	encode predefined relationships between page properties and preprocessing controls

5 Nested-CV Surrogate Model Quality

Table 4: Nested-CV surrogate model quality. Spearman values are reported as mean ± standard deviation across outer folds. The final refit model is selected and fitted on all available pages for saved inference artifacts; reported metrics are out-of-fold nested-CV results.

Setting	Recognizer	Final refit model	MAE	R^2	Global Spearman	Per-image Spearman
Raw	VietOCR	Random Forest	0.021	0.463	0.592 ± 0.123	0.629 ± 0.029
Raw	Tesseract	XGBoost	0.046	0.614	0.674 ± 0.089	0.497 ± 0.027
Raw	EasyOCR	LightGBM	0.039	0.508	0.676 ± 0.062	0.776 ± 0.013
Contour	VietOCR	LightGBM	0.020	0.435	0.522 ± 0.040	0.619 ± 0.037
Contour	Tesseract	Random Forest	0.047	0.576	0.586 ± 0.104	0.527 ± 0.029
Contour	EasyOCR	XGBoost	0.037	0.540	0.668 ± 0.040	0.768 ± 0.031

6 Nested GroupKFold Protocol

7 Nested GroupKFold Evaluation Algorithm

Input: Collected data table; candidate pool Θ; recognizer e; K values $\{3, 5, 10, 15\}$; model-family set $\mathcal{H}$.
Output: Out-of-fold page-level Surrogate K, Random K, FixedTopK, NN-K, FixedBest, NoPre, and sampled-oracle results.

Table 5: Nested GroupKFold evaluation protocol.

Component	Setting	Purpose
Outer evaluation split	5-fold GroupKFold grouped by page image	prevents rows from the same page appearing in both training and test data
Outer test unit	held-out pages, with all 50 candidates kept together	produces out-of-fold page-level evaluation
Inner model selection	5-fold GroupKFold on each outer training split	selects the surrogate model family by validation MAE without using outer test pages
Reported aggregation	held-out page-level results combined across the five outer folds	provides one out-of-fold result per page for CER summaries and statistical tests
Final refit	model family selected and refit on all available pages	produces saved inference artifacts, not reported cross-validation metrics

1. Split pages into five outer GroupKFold folds, keeping all 50 candidate rows from the same page in the same fold.
2. For each outer fold, select $h_e \in \mathcal{H}$ by inner GroupKFold validation MAE on the outer-training pages.
3. Refit the selected model on all outer-training rows and score all $\theta_j \in \Theta$ for each held-out page.
4. Compute Surrogate K, Random K, FixedTopK, NN-K, FixedBest, NoPre, and sampled-oracle references using the definitions in the main manuscript.
5. Pool held-out page-level results across the five outer folds.
6. Compute mean CER, bootstrap confidence intervals, and two-sided Wilcoxon signed-rank tests.

8 Policy-Agnostic Headroom Controls

For each held-out page and budget, the main manuscript defines FixedTopK-to-oracle headroom as $H = F - O$. The coupled outcomes $F - P$, for policy P, share F with H and satisfy $F - P \leq F - O$. We therefore recomputed the correlation for Surrogate, Random, and NN-K and added direct policy-to-policy controls that do not contain FixedTopK. At every budget, all eight dataset/setting/recognizer configurations have positive significant coupled correlations for all three policies. Table 6 gives the per-budget breakdown of the additional controls.

Table 6: Per-budget headroom-control breakdown. Random $r >$ Surrogate r compares the two coupled correlations. The remaining columns count positive Spearman correlations with $p < 0.05$ for direct policy-to-policy gains that do not contain FixedTopK. Each budget contains eight dataset/setting/recognizer configurations.

K	Coupled positive, each policy	Random $r >$ Surrogate r	Random − Surrogate	Random − NN-K	NN-K − Surrogate
3	8/8	7/8	0/8	2/8	0/8
5	8/8	7/8	0/8	1/8	0/8
10	8/8	8/8	0/8	2/8	0/8
15	8/8	8/8	0/8	0/8	0/8

Across all budgets, the coupled relationship is positive and significant in 32/32 configurations for each policy, while Random has the larger correlation in 30/32 Surrogate-versus-Random comparisons. The corresponding direct Random-minus-Surrogate correlation ranges from −0.516 to 0.022, with no positive significant result. These controls show that headroom is a retrospective measure of available improvement in the sampled pool, not evidence specific to learned ranking or an oracle-free policy-selection rule.

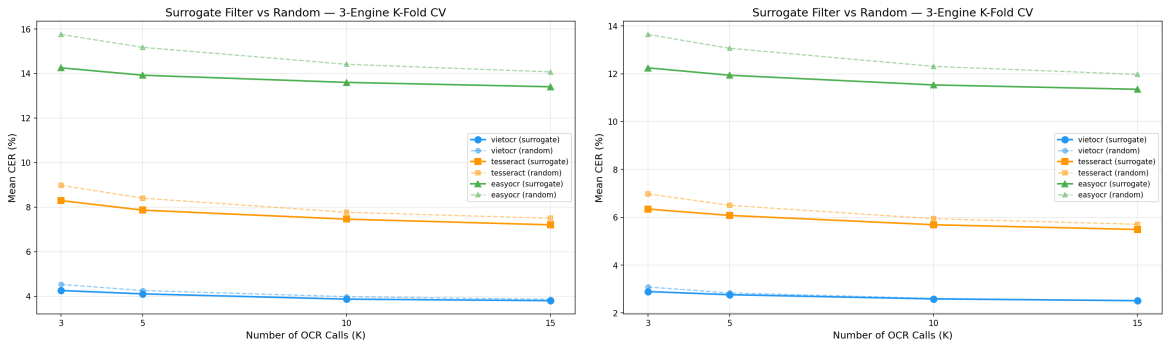


Figure 1: Best-of-K CER comparison for raw and contour-suppressed Vietnamese administrative document settings. These curves correspond to the numerical values reported in the main-manuscript Vietnamese best-of-K table.

9 Vietnamese Best-of-K Curves

10 Feature-Matched Oracle-Free Selector Sensitivity Analysis

The main selector uses continuous surrogate predictions and prediction margins. NN-K and FixedTopK provide only a candidate order. To make these inputs comparable, we replaced the surrogate prediction features by ranks 1, 2, and 3 while preserving the exact surrogate top-3 candidates and OCR outputs. The nested selector was then rerun. Across the eight setting-recognizer combinations, selector CER changed by -0.03 to +0.09 percentage points relative to the full-feature selector. The signs are mixed because the selector is refit, but the small changes are consistent with removing only prediction-related inputs while preserving the exact top-3 candidates and OCR outputs. The table reports paired mean CER differences in percentage points (feature-matched Surrogate minus baseline), percentile bootstrap confidence intervals, and Holm-adjusted two-sided Wilcoxon p-values across the 14 feature-matched comparisons. Negative differences favor the feature-matched Surrogate.

Table 7: Feature-matched oracle-free selector sensitivity analysis. CER values and differences are percentages. Holm p-values are adjusted across all 14 rows in this sensitivity analysis.

Setting	Recognizer	Policy	Matched Surr.	Policy CER	Diff	95% CI	Holm p
Raw	VietOCR	NN-K	4.52	4.73	-0.21	[-0.35, -0.06]	5.89×10^{-4}
Raw	Tesseract	NN-K	9.03	9.31	-0.28	[-0.55, -0.02]	0.433
Raw	EasyOCR	NN-K	14.99	15.33	-0.34	[-0.55, -0.13]	0.0332
Contour	VietOCR	NN-K	3.15	3.27	-0.12	[-0.24, 0.00]	0.458
Contour	Tesseract	NN-K	7.13	7.12	+0.01	[-0.26, 0.29]	1.000
Contour	EasyOCR	NN-K	12.96	13.39	-0.43	[-0.64, -0.22]	0.00125
Raw	VietOCR	FixedTopK	4.52	4.54	-0.02	[-0.15, 0.10]	1.000
Raw	Tesseract	FixedTopK	9.03	9.12	-0.09	[-0.36, 0.19]	1.000
Raw	EasyOCR	FixedTopK	14.99	15.18	-0.19	[-0.35, -0.03]	0.346
Contour	VietOCR	FixedTopK	3.15	3.08	+0.08	[-0.02, 0.17]	0.346
Contour	Tesseract	FixedTopK	7.13	7.05	+0.08	[-0.22, 0.39]	1.000
Contour	EasyOCR	FixedTopK	12.96	13.09	-0.13	[-0.31, 0.05]	0.749
SROIE	Tesseract	FixedTopK	44.27	45.13	-0.86	[-1.20, -0.52]	1.57×10^{-7}
SROIE	EasyOCR	FixedTopK	39.81	39.82	-0.01	[-0.16, 0.14]	1.000

No SROIE NN-K row is included because the stored NN-K shortlist cannot be reconstructed exactly for three EasyOCR pages. The analysis reduces the Surrogate's numerical prediction-feature advantage, but it does not isolate shortlist quality alone because the selector is refit and retains OCR-output features.

11 SROIE Intermediate-Budget Results

12 Contour-Suppression Schematic

Table 8: SROIE receipt-setting intermediate-budget CER results. Values are percentages. These rows complement the main-manuscript SROIE table, which reports $K = 3$ and $K = 15$.

Recognizer	K	Surrogate K	NN-K	FixedTopK	Random K
Tesseract	5	42.34	42.61	43.27	44.79
Tesseract	10	41.39	41.55	42.06	43.02
EasyOCR	5	38.74	38.55	38.87	39.86
EasyOCR	10	38.32	38.27	38.46	38.88

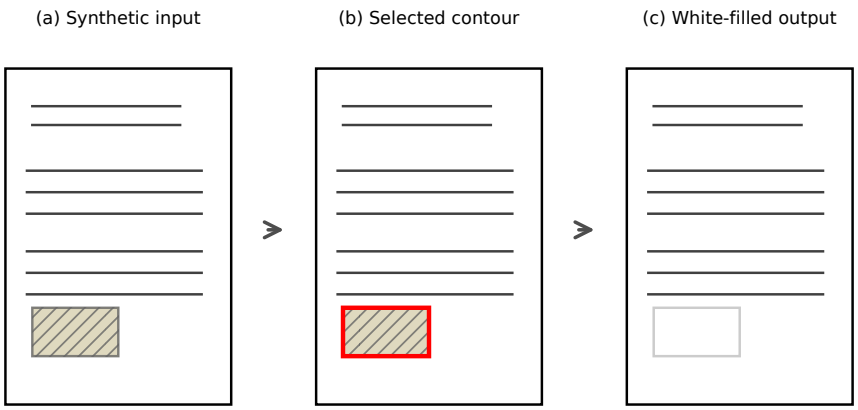


Figure 2: Synthetic illustration of contour suppression. A stamp-like rectangular artifact is identified and white-filled before OCR. The figure is schematic and does not reproduce a real source document.